

Rho estimation options — plain-language guide

For: Charles (edit freely; discuss with Alex)
Last synced: 2026-08-14
Context: Paper Directions 21 — fit homophily knob ρ (rho) on **empirical NCAA rosters**, separate from draft maximum likelihood estimation (MLE)
Companions: - `../Alex_stuff/MLE_basics.md` — Bernoulli draft MLE (λ, t); **no** ρ on fixed rosters - `../Alex_stuff/PD20_K_draws_and_rho_explainer.md` — why ρ and draft likelihood are different layers - `../transcripts/PD21_notes.md` — Alex lock (Aug 14) - `../re_entry/HEROs_and_PASSes/Alex_LG_three_step_briefing.md` — Grandchild assignment (LG) formulas
Locked project choices (Aug 14): - **Assignment engine:** Grandchild assignment (LG) only — not Pass B soft assign - **Data:** empirical NCAA, empirical roster caps, 2011–2021 - **Outputs:** one ρ^* per season + one **longitudinal** ρ^* pooled across seasons - **Seeds:** 50 simulation runs per (ρ , season) to start (increase later; high-performance computing (HPC) when stable) - **Success (stronger):** match sorting index (H_{sort}), global within-team sum of squares (global_wss), and team congestion (L_C) distribution vs empirical; report vs legacy $\rho = 0.5$

What all four options are trying to do

You have **real NCAA men’s basketball data** for 2011–2021. For each season you know:

Input	Meaning
$\hat{A}_i$	Player ability (performance per minute, z-scored within season — hero panel)
Roster caps	How many seats each real team has (stub capacity multiset)
Team labels	Who ended up on which team (observed assignment)

You also have a **simulation rule** for building synthetic rosters: **Grandchild assignment (LG)**. It seats players one at a time. The homophily knob ρ controls how strongly a player prefers teams whose **current** average ability is close to their own.
The fitting problem: find the value of ρ that makes Grandchild assignment (LG) produce rosters that **look like** the real ones — using **team assignment data only**, not draft outcomes.
Important: none of the four options below uses “was this player drafted?” That is a **separate** step (Bernoulli softmax draft MLE for λ, t).

Shared vocabulary

Term	Plain meaning
ρ (rho)	Homophily knob in Grandchild assignment (LG). Higher $\rho \rightarrow$ players sit with more similar teammates. $\rho = 0 \rightarrow$ nearly random seating among open teams.
Sorting index (H_{sort})	After rosters are set: how much do team labels explain ability spread? Near 0 = teams look random; higher = more ability-sorted. Formula: $1 - \frac{\text{within-team sum of squares}}{\text{total sum of squares}}$.
Global within-team sum of squares (global_wss)	Raw measure of spread around team means. Same information as H_{sort} , but not scaled to 0–1.
Team congestion (L_C)	How crowded a team is with “viable” peers (smooth viability formula). Used in scoring later; also a diagnostic for roster structure.
Stub capacity (R_j)	Open seats on team j during seating. Grandchild weights multiply homophily by remaining seats.
Random seed	Grandchild assignment (LG) shuffles player order and randomizes team draws. Same ρ , different seed $\rightarrow$ slightly different rosters.

Option (a): Sorting-index calibration — match one statistic by simulation

Idea in plain language

Grandchild assignment (LG) is **random**. You cannot expect one simulation run to exactly reproduce one real season. Instead:

- Measure the **real** sorting index (H_{sort}) from observed team labels.
- For a trial value of ρ , run Grandchild assignment (LG) many times (50 seeds) with the **same** abilities and roster caps.
- Average the simulated sorting index over those runs.
- Pick the ρ where simulated and empirical sorting indices match.

You are **calibrating** the knob to a **readout**, not writing a full probability formula for the exact roster.

Tiny numeric example (made-up numbers)

Suppose **2015** has 4,000 players and empirical sorting index $H_{\text{sort}}^{\text{emp}} = 0.12$.

Trial ρ	50 simulation runs $\rightarrow$ mean $H_{\text{sort}}^{\text{sim}}$
0.3	≈ 0.07 (too mixed)
0.8	≈ 0.11 (close)
1.5	≈ 0.18 (too sorted)

Pick $\rho^* \approx 0.8$ for 2015.

Repeat for each season $\rightarrow \rho_i^*$ **per season**.

Then pick ρ_{long}^* by minimizing the same mismatch **averaged or summed across all seasons**.

What this is / is not

Is	Is not
Honest about Grandchild assignment (LG) being stochastic	A literal “probability this exact roster happened”
Easy to explain: “we match realized sorting”	Using draft data
Fast to implement (extends existing rho sweep script)	Matching global_wss or L_C unless you add them ($\rightarrow$ option d)

Per-season vs longitudinal

- **Per season** (ρ_i^*): each year gets its own knob — “what homophily matches 2015 NCAA seating?”
- **Longitudinal** (ρ_{long}^*): one knob for 2011–2021 — “one ρ for the whole panel”

Both are natural. They can differ if sorting strength varied by era.

Option (b): Pseudo-likelihood on observed team labels

Idea in plain language

Grandchild assignment (LG) is **sequential** (players arrive one at a time; team centroids update). Real NCAA data gives only the **final** teams — **not** the seating order.

Pseudo-likelihood means: pretend each player’s team choice was a **simple independent decision** based on how close they are to each team’s talent, and ask: *“Under this simplified rule, how plausible are the teams we actually saw?”*

You **do not** re-run the simulator. You look directly at observed rosters.

The simplified rule (conceptual)

For player i , team j , with leave-one-out team mean $\bar{A}_{j,-i}$ (team j ’s average ability **excluding** player i):

$$P(\text{player } i \text{ on team } j) \propto \exp(-\rho \cdot |\hat{A}_i - \bar{A}_{j,-i}|)$$

Multiply (or add log-) these probabilities for **every player’s actual team**. Pick ρ that makes the **observed** assignment score highest.

Tiny example: 6 players, 2 teams

Abilities: players 1–3 are high (10, 9, 8); players 4–6 are low (2, 3, 1).

Observed assignment: - Team A: players 1, 2, 4 $\rightarrow$ team mean ≈ 7.0 - Team B: players 3, 5, 6 $\rightarrow$ team mean ≈ 4.0

Question for player 4 (ability = 2, actually on Team A):

- Leave-one-out mean of Team A (without player 4): $(10 + 9)/2 = 9.5$
- Leave-one-out mean of Team B (without player 4): $(8 + 3 + 1)/3 = 4.0$

Player 4 is **closer** to Team B’s mean. Under **high** ρ , the simplified model says Team A is **surprising** for player 4.

- **Low** ρ : both teams look plausible $\rightarrow$ observed assignment not very surprising.
- **High** ρ : observed assignment looks **less** plausible unless ρ is tuned so distances work out.

Search for the ρ that makes the **whole season’s** observed labels as plausible as possible under this shortcut.

Why “pseudo”?

Because it **ignores**:

- Stub capacity dynamics (R_j shrinking as seats fill)
- Random seating order
- Endogenous centroids updating during the process

It is a **Besag-style** approximation (common in network statistics when the true model is too hard).

When it helps

Good for	Bad for
Fast sanity check: “is there a ρ consistent with labels?”	Being the official Grandchild assignment (LG) generative story
No seeds / no simulation loop	Matching L_C distribution (doesn’t simulate full rosters)
Cross-check against option (a)	Alex briefing if he asked for strict configuration-model MLE

Recommendation: run as **secondary audit**, not primary, unless Alex explicitly wants this as the main estimator.

Option (c): Exact sequential likelihood

Idea in plain language

Replay Grandchild assignment (LG) **exactly** as coded:

1. Shuffle player order (random permutation).
2. For each player in that order, compute weights $R_j \exp(-\rho|\hat{A}_i - \mu_j|)$ over open teams.
3. Normalize to probabilities, draw a team, update centroid μ_j .
4. Multiply the probability of **each step’s choice** along that order.

For **empirical** data, you don’t know which permutation NCAA “used.” So you must either:

- **Average** over many random orders (expensive), or
- **Sum** over all permutations (impossible at NCAA scale).

Tiny example: 3 players, 2 teams (capacities 2 and 1)

Players: A=10, B=5, C=1. Team 1 has 2 seats, Team 2 has 1 seat.

One possible order: B, then C, then A.

- **Step 1 (B=5):** both teams open, centroids at league mean → B lands Team 1 with some probability (e.g. 0.55).
- **Step 2 (C=1):** Team 1 centroid updates → C lands somewhere with some probability.
- **Step 3 (A=10):** only open seats left; often forced or nearly forced.

Likelihood for **this order** = product of the three step probabilities.

Different order (A first, then B, then C) → **different** product, because centroids evolve differently.

For empirical fit: average likelihood over many orders, or treat order as random and integrate.

Tradeoffs

Pros	Cons
Closest to “true” Grandchild assignment (LG) generative model	Slow and fiddly
Principled if you handle order correctly	At NCAA scale (thousands of players), brutal without serious compute
	Still doesn’t uniquely identify order for real data

Verdict for v1: skip unless Alex insists on full generative MLE. Options (a) and (d) capture the same ρ intent with far less pain.

Option (d): Multi-moment simulation matching — “stronger” criterion

Idea in plain language

Same engine as option (a) — run Grandchild assignment (LG) many times per ρ — but match **several** empirical summaries at once, not just sorting index (H_{sort}):

1. **Sorting index (H_{sort})** — how sorted are teams?
2. **Global within-team sum of squares (global_wss)** — raw spread around team means
3. **Team congestion (L_C) distribution** — histogram of team-level congestion (see `grandchild_empirical_lc_compare.py`)

Define a **combined loss**, e.g.:

$$\text{loss}(\rho) = w_1 \cdot |H_{\text{sort}}^{\text{sim}} - H_{\text{sort}}^{\text{emp}}| + w_2 \cdot |\text{global_wss}^{\text{sim}} - \text{global_wss}^{\text{emp}}| + w_3 \cdot \text{distance}(L_C^{\text{sim}}, L_C^{\text{emp}})$$

(simulated values = mean over 50 seeds)

Pick ρ that minimizes total loss — **per season** and **longitudinally**.

Tiny example (made-up)

Empirical 2015 targets:

- $H_{\text{sort}}^{\text{emp}} = 0.12$
- $\text{global_wss}^{\text{emp}} = 8200$
- L_C distribution: peak near 0.45

At $\rho = 0.5$ (legacy default), 50 sim runs might give:

- $H_{\text{sort}}^{\text{sim}} = 0.09 \rightarrow$ error 0.03

- $\text{global_wss}^{\text{sim}} = 9100 \rightarrow \text{error } 900$
- L_C peak at 0.38 $\rightarrow$ histogram distance 0.05

At $\rho = 0.9$:

- $H_{\text{sort}}^{\text{sim}} = 0.12 \rightarrow \text{error } 0.00$
- $\text{global_wss}^{\text{sim}} = 8500 \rightarrow \text{error } 300$
- L_C peak at 0.43 $\rightarrow$ distance 0.02

Combined loss might favor $\rho \approx 0.9$ even though $\rho = 0.5$ was “close enough” on sorting alone.

Why this matches the “stronger” success criterion

- Option (a) alone: “get sorting index right.”
- Option (d): “get sorting **and** within-team spread **and** congestion shape right” — stronger claim that Grandchild assignment (LG) reproduces **structure**, not one number.

Recommended v1 **primary**, with option (b) as optional audit.

Side-by-side comparison

	(a) Sorting calibration	(b) Pseudo-likelihood	(c) Exact sequential LL	(d) Multi-moment sim match
Uses simulator?	Yes, many seeds	No	Yes, many orders	Yes, many seeds
Uses observed team labels directly?	Only for target stats	Yes, in formula	Yes, in path	Only for target stats
Handles randomness?	Average over seeds	N/A	Average over orders	Average over seeds
Matches H_{sort} ?	Yes	Indirectly	Yes	Yes
Matches $\text{global_wss} / L_C$?	Not unless added	No	Can in principle	Yes
Implementation effort	Low	Low	High	Medium
Alex story	“Calibrate ρ to sorting”	“Approximate label likelihood”	“Full generative MLE”	“Calibrate ρ to roster geometry”

How this connects to implementation choices

Choice	Meaning
50 seeds	For (a) and (d), each (ρ, season) runs Grandchild assignment (LG) 50 times; report mean $\pm$ spread
One ρ per season + longitudinal ρ	Grid search inside each season; then one pooled objective across 2011–2021
$\rho = 0.5$ reference	Always report the same three metrics at legacy default
HPC / Rivanna	Not required to build the script; parallelize $(\text{season}, \rho, \text{seed})$ when seeds or grid fineness increase

Planned outputs (when coded): `HER0s_and_PASSes/pd21_rho/` — script, JSON, plots.

Suggested path forward

Role	Option
Primary	(d) — multi-moment simulation matching
Audit	(b) — pseudo-likelihood, reported alongside

One-liner for Alex:

“We calibrate Grandchild assignment’s ρ so simulated rosters match empirical sorting, within-team spread, and L_C shape — per season and pooled — not from draft likelihood. Draft MLE for λ and t stays on fixed empirical teams.”

Charles runs PDF when ready:

```
./scripts/convert_single_md_to_pdf.sh 3-Master_Plan/Alex_notes/rho_est_options_for_dummies.md
```